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A Likelihood Ratio Approach to Sequential Change Point Detection for a General Class of Parameters

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ABSTRACT

In this article, we propose a new approach for sequential monitoring of a general class of parameters of a d-dimensional time series, which can be estimated by approximately linear functionals of the empirical distribution function. We consider a closed-end method, which is motivated by the likelihood ratio test principle and compare the new method with two alternative procedures. We also incorporate self-normalization such that estimation of the long-run variance is not necessary. We prove that for a large class of testing problems the new detection scheme has asymptotic level α and is consistent. The asymptotic theory is illustrated for the important cases of monitoring a change in the mean, variance, and correlation. By means of a simulation study it is demonstrated that the new test performs better than the currently available procedures for these problems. Finally, the methodology is illustrated by a small data example investigating index prices from the dot-com bubble. Supplementary materials for this article are available online.

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1. Introduction

An important problem in statistical modeling of time series is the problem of testing for structural stability as changes in the data generating process may have a substantial impact on statistical inference developed under the assumption of stationarity. Because of its importance there exists a large amount of literature, which develops tests for structural breaks in various models and we refer to Aue and Horváth (2013) and Jandhyala et al. (2013) for more recent reviews of the literature. There are essentially two ways how the problem of change point analysis is addressed. A large portion of the literature discusses a-posteriori change point analysis, where the focus is on the detection of structural breaks given a historical dataset (see Davis, Huang, and Yao 1995; Csörgö and Horváth 1997; Aue, Horváth, and Reimherr 2009; Jirak 2015, among many others). On the other hand, in many applications, such as engineering, medicine, or risk management data arrive steadily and therefore several authors have addressed the problem of sequentially monitoring changes in a parameter. Page (1954), Hinkley (1971), Moustakides (1986), Nikiforov (1987), and Lai (1995), among others developed detection schemes for models with an infinite time horizon. These methods always stop and have been partly summarized by the term "statistical process control" (SPC). SPC-statistics, for example, CUSUM- or Shewart-Controlcharts, have then been traditionally compared by the average run length (ARL). Our approach differs from this literature substantially as we work under the paradigm introduced by Chu, Stinchcombe, and White (1996). These authors proposed an alternative monitoring ansatz in the context of testing the structural stability of the parameters in a linear model, which on the one hand allows to control (asymptotically) the Type I error (if no changes occur) and on the other hand provides the possibility of power analysis. Horváth et al. (2004) and Fremdt (2014) extended this approach for linear models with infinite time horizon, while Aue et al. (2012), Wied and Galeano (2013), and Pape, Wied, and Galeano (2016) developed monitoring procedures for changes in a capital asset pricing model, correlation, and variance under the assumption of a finite time horizon.

In this article, we propose a general alternative sequential test in this context, which is applicable for change point analysis of a p-dimensional parameter of a d-dimensional time series if a historical dataset from a stable phase is available and then data arrive consecutively. Our approach differs from the current methodology as it is motivated by a likelihood ratio test for a structural change. To be precise, Wied and Galeano (2013) proposed to compare estimates from the historical dataset, say $X_1, \ldots, X_m$, with estimators from the sequentially observed data $X_{m+1}, \ldots, X_{m+k}$ (see also Chu, Stinchcombe, and White 1996; Horváth et al. 2004; Aue et al. 2012; Pape, Wied, and Galeano 2016), while Fremdt (2014) and Kirch and Weber (2018) suggested to compare the estimate from the historical dataset with estimates from the sequentially observed data $X_{m+j+1}, \ldots, X_{m+k}$ (for all $j = 0, \ldots, k-1$). In contrast, motivated by the likelihood ratio principle, our approach sequentially compares the estimates from the samples $X_1, \ldots, X_{m+j}$ and $X_{m+j+1}, \ldots, X_{m+k}$ (for all $j = 0, \ldots,$ k-1).

Moreover, we also propose a self-normalized test, which avoids the problem of estimating the long-run variance. While

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the concept of self-normalization has been studied intensively for a-posteriori change point analysis (see Shao and Zhang 2010; Shao 2015; Zhang and Lavitas 2018, among many others), to our best knowledge self-normalization has not been studied in the context of sequential monitoring.

The statistical model and the change point problem for a general parameter of the marginal distribution are introduced in Section 2, where we also provide the motivation for the statistic used in the sequential scheme (see Example 2.1) and a discussion of the alternative methods. The asymptotic properties of the new monitoring scheme are investigated in Section 3. In particular we prove a result, which allows to control (asymptotically) the probability of indicating a change in the parameter although there is in fact structural stability (Type I error). Moreover, we also show that the new test is consistent and investigate the concept of self-normalization in this context. These asymptotic considerations require several assumptions, which are stated in the general context and verified in Section 4 for the case of monitoring changes in the mean, variance matrix and quantizes. In Section 5, the finite sample properties of the new procedure are investigated by means of a simulation study, and we also demonstrate the (empirical) superiority of our approach. Here we also illustrate our approach by a small real data example investigating the log-returns of NASDAQ and S&P 500 during the dot-com bubble. Finally, Section 6 gives a brief conclusion and an outlook defining subjects for future research. All proofs are deferred to an online supplement.

2. Sequential Change Point Testing

Consider a d-dimensional time series $\{X_t\}_{t\in\mathbb{Z}}$, where X_t has distribution function F_t, and denote by $\theta_t = \theta(F_t)$ a p-dimensional parameter of interest of the distribution of F_t. We are taking the sequential point of view and assume that there exists a historical period of length, say $m \in \mathbb{N}$, such that the process is stable in the sense

$$\theta_1 = \theta_2 = \cdots = \theta_m. \tag{1}$$

We are interested to monitor if the parameter θ_{m+k} changes in the future $m + k \geq m + 1$. The sequence $X_1, \ldots, X_m$ is usually referred to as a historical or initial training dataset (see, e.g., Chu, Stinchcombe, and White 1996; Horváth et al. 2004; Wied and Galeano 2013; Kirch and Weber 2018, among many others). Based on this stretch of "stable" observations a sequential procedure should be conducted to test the hypotheses

$$H_0 : \theta_1 = \cdots = \theta_m = \theta_{m+1} = \theta_{m+2} = \cdots, \tag{2}$$

against the alternative that the parameter $\theta_{m+k^\star}$ changes for some $k^\star \geq 1$, that is

$$H_1 : \exists k^\star \in \mathbb{N} : \theta_1 = \cdots = \theta_{m+k^\star-1} \neq \theta_{m+k^\star} = \theta_{m+k^\star+1} = \cdots. \tag{3}$$

To motivate our approach in particular the test statistic used in the detection scheme, which will be used in the proposed sequential test, we begin with a very simple example of a change in the mean.

Example 2.1. Consider a sequence $\{X_t\}_{t\in\mathbb{Z}}$ of independent, d-dimensional normal distributed random variables with (positive definite) variance matrix Σ and mean vectors

$$\mu_t = \mathbb{E}[X_t] = \theta(F_t) = \int_{\mathbb{R}^d} x dF_t(x) \in \mathbb{R}^d, \quad t = 1, 2, \ldots. \tag{4}$$

During the monitoring procedure, we propose to successively test the hypotheses

$$\begin{aligned} H_0 :\ & \mu_1 = \cdots = \mu_m = \mu_{m+1} = \cdots = \mu_{m+k}, \text{ versus} \\ H_A^{(k)} : \exists\, j \in \{0, \ldots, k-1\} :\ & \mu_1 = \cdots = \mu_{m+j} \neq \mu_{m+j+1} \\ & = \cdots = \mu_{m+k} \end{aligned} \tag{5}$$

based on the sample $X_1, \ldots, X_{m+k}$. Under the assumptions made in this example we can easily derive the likelihood ratio

$$\Lambda_m(k) = \frac{\sup\limits_{\mu\in\mathbb{R}^d} \prod\limits_{t=1}^{m+k} f(X_t, \mu)}{\sup\limits_{\substack{j\in\{0,\ldots,k-1\} \\ \mu^{(1)},\mu^{(2)}\in\mathbb{R}^d}} \prod\limits_{t=1}^{m+j} f(X_t, \mu^{(1)}) \cdot \prod\limits_{t=m+j+1}^{m+k} f(X_t, \mu^{(2)})},$$

where $f(\cdot, \mu)$ denotes the density of a normal distribution with mean μ and variance matrix Σ (note that the first m observations are assumed to be mean-stable). A careful calculation now proves the identity

$$\begin{aligned} -2\log\big(\Lambda_m(k)\big) &= \max_{j=0}^{k-1} \frac{(m+j)(k-j)}{m+k} \big(\hat\mu_1^{m+j} - \hat\mu_{m+j+1}^{m+k}\big)^\top \\ &\quad \times \Sigma^{-1}\big(\hat\mu_1^{m+j} - \hat\mu_{m+j+1}^{m+k}\big) \\ &= \max_{j=0}^{k-1} \frac{(m+k)(m+j)}{(k-j)} \big(\hat\mu_1^{m+j} - \hat\mu_1^{m+k}\big)^\top \\ &\quad \times \Sigma^{-1}\big(\hat\mu_1^{m+j} - \hat\mu_1^{m+k}\big), \end{aligned} \tag{6}$$

where $v^\top$ denotes the transposed of the vector v (usually considered as a column vector) and

$$\hat\mu_i^j = \frac{1}{j-i+1} \sum_{t=i}^{j} X_t$$

is the mean of the observations $X_i, \ldots, X_j$. Consequently the null hypothesis $H_0^{(k)}$ should be rejected in favor of the alternative $H_A^{(k)}$ for large values of the statistic

$$\max_{j=0}^{k-1} (m+j)(k-j)\big(\hat\mu_1^{m+j} - \hat\mu_{m+j+1}^{m+k}\big)^\top \Sigma^{-1}\big(\hat\mu_1^{m+j} - \hat\mu_{m+j+1}^{m+k}\big). \tag{7}$$

However, as pointed out in Csörgö and Horváth (1997), the asymptotic properties of a likelihood ratio type statistic of the type (7) are difficult to study. For this reason, we propose to use a weighted version of (7) and consider the statistic

$$\begin{aligned} &\max_{j=0}^{k-1} (m+j)^2(k-j)^2\big(\hat\mu_1^{m+j} - \hat\mu_{m+j+1}^{m+k}\big)^\top \\ &\qquad \times \Sigma^{-1}\big(\hat\mu_1^{m+j} - \hat\mu_{m+j+1}^{m+k}\big) \\ &= \max_{j=0}^{k-1} \Big((k-j)\sum_{t=1}^{m+j} X_t - (m+j)\sum_{t=m+j+1}^{m+k} X_t\Big) \\ &\qquad \times \Sigma^{-1}\Big((k-j)\sum_{t=1}^{m+j} X_t - (m+j)\sum_{t=m+j+1}^{m+k} X_t\Big), \end{aligned} \tag{8}$$

for which (after appropriate normalization) weak convergence of a corresponding sequential empirical process can be established. Note that the right-hand side in (8) corresponds to the well-known CUSUM statistic, which has become a standard tool for change point detection in a retrospective setting.

Motivated by the previous example we propose to use the statistic

$$\hat{D}_m(k) = m^{-3} \max_{j=0}^{k-1} (m+j)^2 (k-j)^2 \big(\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\big)^\top \times \hat{\Sigma}_m^{-1} \big(\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\big), \tag{9}$$

for monitoring changes in the parameter θ_j, where $\hat{\theta}_i^j = \theta(\hat{F}_i^j)$ denotes the estimator obtained from the empirical distribution function

$$\hat{F}_i^j(z) = \frac{1}{j-i+1} \sum_{t=i}^{j} I\{X_t \le z\} \tag{10}$$

of the observations $X_i, \ldots, X_j$ and the matrix $\hat{\Sigma}_m$ corresponds to an estimator of a long-run variance based on $X_1, \ldots, X_m$. The scaling by m^{-3} will be necessary to obtain weak convergence in the sequel.

We use the sequence $\{\hat{D}_m(k)\}_{k\in\mathbb{N}}$ in combination with an increasing threshold function $w(\cdot)$ as a monitoring scheme. More precisely, let $T \in \mathbb{R}_+$ denote a constant factor (with $Tm \in \mathbb{N}$) defining the window of monitoring and let $w : [0, T] \to \mathbb{R}_+$ be increasing. Then we reject the null hypothesis in (2) at the first time $k \in \{1, 2, \ldots, Tm\}$ for which the detector $\hat{D}_m(k)$ exceeds the so-called critical curve/boundary, that is

$$\hat{D}_m(k) > C_\alpha w(k/m), \tag{11}$$

where the constant C_α has to be chosen such that the test has asymptotic level α, which is

$$\limsup_{m\to\infty} \mathbb{P}_{H_0}\left(\max_{k=1}^{mT} \frac{\hat{D}_m(k)}{w(k/m)} > C_\alpha \right) \le \alpha. \tag{12}$$

The definition above yields a stopping rule defined by

$$\tau_m = \inf\left\{ 1 \le k \le Tm \mid \hat{D}_m(k) > C_\alpha w(k/m) \right\}$$

with the convention that $\tau_m = \infty$ if the set $\{1 \le k \le Tm \mid \hat{D}_m(k) > C_\alpha w(k/m)\}$ is empty. We will call the procedure/test consistent if

$$\lim_{m\to\infty} \mathbb{P}_{H_1}\left(\max_{k=1}^{mT} \frac{\hat{D}_m(k)}{w(k/m)} > C_\alpha \right) = \lim_{m\to\infty} \mathbb{P}_{H_1}(\tau_m < \infty) = 1. \tag{13}$$

Following Aue et al. (2012) we call this procedure a closed-end method, because monitoring is only performed in the interval $[m+1, mT]$.

To our best knowledge, the detection scheme defined by (11) has not been considered in the literature so far. However, our approach is related to the work of Chu, Stinchcombe, and White (1996), Horváth et al. (2004), Aue et al. (2012), Wied and Galeano (2013), and Pape, Wied, and Galeano (2016), who investigated sequential monitoring schemes for various parameters (such as the correlation, the variance, or the parameters of the capital asset pricing model). In the general situation considered in this section their approach uses

$$\hat{Q}_m(k) = \frac{k^2}{m} \big(\hat{\theta}_1^m - \hat{\theta}_{m+1}^{m+k}\big)^\top \hat{\Sigma}_m^{-1} \big(\hat{\theta}_1^m - \hat{\theta}_{m+1}^{m+k}\big) \tag{14}$$

as a basic statistic in the sequential procedure. Note that a sequential scheme based on this statistic measures the differences between the estimator $\hat{\theta}_1^m$ from the initial data and the estimator $\hat{\theta}_{m+1}^{m+k}$ from all observations excluding the training sample. As a consequence—in particular in the case of a rather late change—the estimator $\hat{\theta}_{m+1}^{m+k}$ may be corrupted by observations before the change point, which might lead to a loss of power. Another related procedure uses the statistic

$$\hat{P}_m(k) = \max_{j=0}^{k-1} \frac{(k-j)^2}{m} \big(\hat{\theta}_1^m - \hat{\theta}_{m+j+1}^{m+k}\big)^\top \hat{\Sigma}_m^{-1} \big(\hat{\theta}_1^m - \hat{\theta}_{m+j+1}^{m+k}\big) \tag{15}$$

and was recently suggested by Fremdt (2014) and reconsidered by Kirch and Weber (2018). These authors compare the estimate from the data $X_1, \ldots, X_m$ with estimates from the data $X_{m+j+1}, \ldots, X_{m+k}$ (for different values of j). This method may lead to a loss in power in problems with a small sample of historical data and a rather late change point. In contrast our approach compares the estimates of the parameters before and after all potential positions of a change point $j \in \{m+1, \ldots, m+k\}$.

In this article, we argue that the performance of the change point tests can be improved by replacing $\hat{\theta}_1^m$ by $\hat{\theta}_1^{m+j}$ inside the maximum, which would directly lead to a scheme of the form (9). Here, we would already like to point to our simulation study in Section 5, which contains many cases where a sequential detection scheme based on the statistic $\hat{D}_m$ outperforms schemes based on $\hat{Q}_m$ or $\hat{P}_m$.

3. Asymptotic Properties

In the subsequent discussion we use the following notation. We denote by $\ell^\infty(V_1, V_2)$ the space of all bounded functions $f : V_1 \to V_2$ equipped with sup-norm, where V_1, V_2 are normed linear spaces. The symbols $\xrightarrow{\mathbb{P}}$ and $\xrightarrow{\mathcal{D}}$ mean convergence in probability and weak convergence (in the space under consideration), respectively. The process $\{W(s)\}_{s\in[0,T+1]}$ will usually represent a standard p-dimensional Brownian motion. For a vector $v \in \mathbb{R}^d$, we denote by $|v| = \left(\sum_{i=1}^d v_i^2\right)^{1/2}$ its Euclidean norm.

3.1. Weak Convergence

Throughout this article we denote by $\theta = \theta(F)$ a p-dimensional functional of the d-dimensional distribution function F and define its influence function (assuming its existence) by

$$\mathcal{IF}(x, F, \theta) = \lim_{\varepsilon \searrow 0} \frac{\theta((1-\varepsilon)F + \varepsilon\delta_x) - \theta(F)}{\varepsilon}, \tag{16}$$

where $\delta_x(z) = I\{x \le z\}$ is the distribution function of the Dirac measure at the point $x \in \mathbb{R}^d$ and the inequality in the indicator is understood component-wise. Throughout this section we make the following assumptions, which will be verified for several important examples in Section 4 (see also Shao and Zhang 2010 for similar regularity conditions).

Assumption 3.1. Under the null hypothesis (2) we assume that the times series $\{X_t\}_{t\in\mathbb{N}}$ is strictly stationary with $\mathbb{E}[\mathcal{IF}(X_1, F, \theta)] = 0$ and that the weak convergence

$$\frac{1}{\sqrt{m}} \sum_{t=1}^{\lfloor ms \rfloor} \mathcal{IF}(X_t, F, \theta) \overset{\mathcal{D}}{\Longrightarrow} \sqrt{\Sigma_F} W(s), \tag{17}$$

holds in the space $\ell^\infty([0, T+1], \mathbb{R}^p)$ as $m \to \infty$, where the long-run variance matrix is defined by (assuming convergence of the series)

$$\Sigma_F = \sum_{t\in\mathbb{Z}} \operatorname{cov}\big(\mathcal{IF}(X_0, F, \theta),\ \mathcal{IF}(X_t, F, \theta)\big) \in \mathbb{R}^{p\times p} \tag{18}$$

and $\{W(s)\}_{s\in[0,T+1]}$ is a p-dimensional (standard) Brownian motion.

Assumption 3.2. The remainder terms

$$R_{i,j} = \hat{\theta}_i^j - \theta(F) - \frac{1}{j-i+1} \sum_{t=i}^{j} \mathcal{IF}(X_t, F, \theta) \tag{19}$$

in the linearization of $\hat{\theta}_i^j - \theta(F)$ satisfy

$$\sup_{1\le i<j\le n} (j-i+1)|R_{i,j}| = o_{\mathbb{P}}(n^{1/2}). \tag{20}$$

For the statement of our first result we introduce the notations (throughout this article we use the convention $\hat{\theta}_z^u = 0$, whenever $z > u$)

$$\tilde{\mathbb{U}}(\ell, z, u) := (u-z)(z-\ell)\big(\hat{\theta}_{\ell+1}^z - \hat{\theta}_{z+1}^u\big), \tag{21}$$

$$\mathbb{U}(z, u) := \tilde{\mathbb{U}}(0, z, u) = (u-z)z\big(\hat{\theta}_1^z - \hat{\theta}_{z+1}^u\big), \tag{22}$$

and denote by

$$\Delta_2 = \{(s,t) \in [0, T+1]^2 \mid s \le t\}, \tag{23}$$

$$\Delta_3 = \big\{(r,s,t) \in [0, T+1]^3 \mid r \le s \le t\big\}. \tag{24}$$

The two-dimensional triangle and the three-dimensional oblique pyramid in $[0, T+1]^2$ and $[0, T+1]^3$, respectively.

Theorem 3.1. Let Assumptions 3.1 and 3.2 be satisfied. If the null hypothesis in (2) holds, then as $m \to \infty$

$$\big\{m^{-3/2}\tilde{\mathbb{U}}(\lfloor mr \rfloor, \lfloor ms \rfloor, \lfloor mt \rfloor)\big\}_{(r,s,t)\in\Delta_3} \overset{\mathcal{D}}{\Longrightarrow} \Sigma_F^{1/2}\big\{B(s,t) + B(r,s) - B(r,t)\big\}_{(r,s,t)\in\Delta_3} \tag{25}$$

in the space $\ell^\infty(\Delta_3, \mathbb{R}^p)$, where the process $\{B(s,t)\}_{(s,t)\in\Delta_2}$ is defined by

$$B(s,t) = tW(s) - sW(t), \quad (s,t) \in \Delta_2, \tag{26}$$

and $\{W(s)\}_{s\in[0,T+1]}$ denotes a p-dimensional Brownian motion on the interval $[0, T+1]$.

Remark 3.1. As a by-product of Theorem 3.1 and the representation (22) we obtain the weak convergence of the double-indexed CUSUM-process (22), that is ($m \to \infty$)

$$\big\{m^{-3/2} \cdot \mathbb{U}(\lfloor ms \rfloor, \lfloor mt \rfloor)\big\}_{(s,t)\in\Delta_2} \overset{\mathcal{D}}{\Longrightarrow} \{\Sigma_F^{1/2} B(s,t)\}_{(s,t)\in\Delta_2}, \tag{27}$$

where Δ_2 denotes the two-dimensional triangle in $[0, T+1]^2$ and the process B is defined in (26). In particular the covariance structure of the process B is given by

$$\operatorname{cov}\big(B(s_1,t_1), B(s_2,t_2)\big) = t_1t_2(s_1 \wedge s_2) - t_1s_2(s_1 \wedge t_2) - s_1t_2(t_1 \wedge s_2) + s_1s_2(t_1 \wedge t_2).$$

Consequently, the process $\{B(s,t)\}_{(s,t)\in\Delta_2}$ can be considered as a natural extension of the standard Brownian bridge as for fixed t the process $\{B(s,t)\}_{s\in[0,t]}$ is a Brownian bridge on the interval $[0, t]$.

Observing the definition (22) the statistic (9) allows the representation

$$\hat{D}_m(k) = m^{-3} \max_{j=0}^{k-1} |\mathbb{U}^\top(m+j, m+k)\hat{\Sigma}_m^{-1}\mathbb{U}(m+j, m+k)|, \tag{28}$$

and we obtain the following corollary as a consequence of Theorem 3.1.

Corollary 3.1. Let the assumptions of Theorem 3.1 be satisfied. If the null hypothesis in (2) holds, and $\hat{\Sigma}_m$ denotes a consistent, nonsingular estimator of the long-run variance Σ_F defined in (18), then as $m \to \infty$

$$\max_{k=1}^{Tm} \frac{\hat{D}_m(k)}{w(k/m)} \overset{\mathcal{D}}{\Longrightarrow} \sup_{t\in[1,T+1]} \sup_{s\in[1,t]} \frac{B(s,t)^\top B(s,t)}{w(t-1)} = \sup_{t\in[0,T]} \sup_{s\in[0,t]} \frac{B(s+1,t+1)^\top B(s+1,t+1)}{w(t)},$$

for any threshold function $w : [0, T] \to \mathbb{R}^+$, which is increasing.

By the result in Corollary 3.1 it is reasonable to choose C_α for a given level α and threshold function $w(\cdot)$, such that

$$\mathbb{P}\left(\sup_{t\in[1,T+1]} \sup_{s\in[1,t]} \frac{B(s,t)^\top B(s,t)}{w(t-1)} > C_\alpha\right) = \alpha \tag{29}$$

and to reject the null hypothesis H_0 in (2) at time k, if

$$\hat{D}_m(k) > C_\alpha w(k/m). \tag{30}$$

By Corollary 3.1 this test has asymptotic level α, that is

$$\lim_{m\to\infty} \mathbb{P}_{H_0}\left(\max_{k=1}^{Tm} \frac{\hat{D}_m(k)}{w(k/m)} > C_\alpha\right) = \alpha$$

if the assumptions of Theorem 3.1 hold and w and C_α satisfy (29). The choice of $w(\cdot)$ has been investigated by several authors (see Chu, Stinchcombe, and White 1996; Aue, Horváth, and Reimherr 2009; Wied and Galeano 2013, among others) and we will compare different options by means of a simulation study in Section 5. Note that one can take any function (which is increasing and bounded from below by a positive constant) and then choose the constant C_α as the corresponding quantile of the supremum of the limit process such that (29) is fulfilled. Next we discuss the consistency of the monitoring scheme (30). For this purpose, we consider the alternative hypothesis in (3), where the location of the change point is increasing with the length of the training sample, that is $m + k^* = \lfloor mc \rfloor$ for some

$1 < c < T+1$. Recalling the definition of $\hat{D}_m(k)$ in (9) and observing the inequality

$$\max_{k=1}^{Tm} \hat{D}_m(k) \geq \frac{\lfloor mc \rfloor^2 \big(T(m+1) - \lfloor mc \rfloor\big)^2}{m^3} \times \frac{\big(\hat{\theta}_1^{\lfloor mc \rfloor} - \hat{\theta}_{\lfloor mc \rfloor+1}^{T(m+1)}\big)^\top \hat{\Sigma}_m^{-1} \big(\hat{\theta}_1^{\lfloor mc \rfloor} - \hat{\theta}_{\lfloor mc \rfloor+1}^{T(m+1)}\big)}{w(T)}$$

it is intuitively clear that the statistic $\max_{k=1}^{Tm} \hat{D}_m(k)$ converges to infinity, provided that $\hat{\theta}_1^{\lfloor mc \rfloor}$ and $\hat{\theta}_{\lfloor mc \rfloor+1}^{T(m+1)}$ are consistent estimates of the parameter θ before and after the change point $m + k^* = \lfloor mc \rfloor$ and $\hat{\Sigma}_m$ converges to a positive definite $p \times p$ matrix. The following Theorem 3.2 makes these heuristic arguments more precise. Its proof requires several assumptions, which are stated first. The result might be even correct under slightly weaker assumptions. However, in the form stated below we can also prove consistency of a sequential scheme based on a self-normalized version of $\hat{D}_m(k)$ (see Theorem 3.3 in Section 3.2).

Assumption 3.3. If the alternative hypothesis H_1 defined in (3) holds we assume that the change occurs at position $m + k^* = \lfloor mc \rfloor$ for some $c \in (1, T+1)$. Moreover, let $\{Z_t(1)\}_{t\in\mathbb{Z}}$ and $\{Z_t(2)\}_{t\in\mathbb{Z}}$ denote (strictly) stationary $\mathbb{R}^d$-valued processes with marginal distribution functions $F^{(1)}$ and $F^{(2)}$, respectively, such that

$$\theta(F^{(1)}) \neq \theta(F^{(2)}),$$

and that for each $m \in \mathbb{N}$

$$(X_1, X_2, \ldots, X_{\lfloor mc \rfloor}) \stackrel{\mathcal{D}}{=} (Z_1(1), \ldots, Z_{\lfloor mc \rfloor}(1)), \tag{31}$$

$$(X_{\lfloor mc \rfloor+1}, \ldots, X_{\lfloor mT \rfloor}) \stackrel{\mathcal{D}}{=} (Z_{\lfloor mc \rfloor+1}(2), \ldots, Z_{\lfloor mT \rfloor}(2)). \tag{32}$$

Note, that formally the process $\{X_t\}_{t=1,\ldots\lfloor mT \rfloor}$ is a triangular array, that is $X_t = X_{m,t}$, but we do not reflect this in our notation. Further, we assume that there exist two (standard) Brownian motions W_1 and W_2 such that the joint weak convergence

$$\begin{pmatrix} \Big\{\frac{1}{\sqrt{m}} \sum_{t=1}^{\lfloor ms \rfloor} \mathcal{IF}(Z_t(1), F^{(1)}, \theta)\Big\}_{s\in[0,c]} \\ \Big\{\frac{1}{\sqrt{m}} \sum_{t=\lfloor mc \rfloor+1}^{\lfloor ms \rfloor} \mathcal{IF}(Z_t(2), F^{(2)}, \theta)\Big\}_{s\in[c,T+1]} \end{pmatrix} \stackrel{\mathcal{D}}{\Longrightarrow} \begin{pmatrix} \big\{\sqrt{\Sigma_{F^{(1)}}} W_1(s)\big\}_{s\in[0,c]} \\ \big\{\sqrt{\Sigma_{F^{(2)}}}\big(W_2(s) - W_2(c)\big)\big\}_{s\in[c,T+1]} \end{pmatrix}$$

holds, where $\Sigma_{F^{(1)}}$ and $\Sigma_{F^{(2)}}$ denote positive definite matrices defined in the same way as (18), that is

$$\Sigma_{F^{(\ell)}} = \sum_{t\in\mathbb{Z}} \operatorname{cov}\big(\mathcal{IF}(Z_0(\ell), F^{(\ell)}, \theta), \mathcal{IF}(Z_t(\ell), F^{(\ell)}, \theta)\big) \in \mathbb{R}^{p\times p}, \quad \ell = 1, 2 \tag{33}$$

and both phases in (31) and (32) fulfill Assumption 3.2 for the corresponding expansion.

Remark 3.2.

(a) The interpretation of Assumption 3.3 is as follows: There exist two regimes and the process under consideration switches from one regime to the other.

(b) The assumption of two stationary phases before and after the change point is commonly made in the literature to analyze change point tests under the alternative (see, e.g., Aue, Horváth, and Reimherr 2009; Dette and Wied 2016; Kirch and Weber 2018, among others). Note, that we do not assume that the two limiting processes W_1 and W_2 are independent.

(c) Often Assumption 3.3 is directly implied by the underlying change point problem. For example, in the situation of the mean vector introduced in (2.1) it is usually assumed that $X_t = \mu_t + \varepsilon_t$, where $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ is a stationary process and $\mu_t = \mu^{(1)}$ if $t \leq \lfloor mc \rfloor$ and $\mu_t = \mu^{(2)}$ if $t \geq \lfloor mc \rfloor + 1$. In this case Assumption 3.3 is obviously satisfied. Further examples are discussed in Section 4.

Theorem 3.2. Let Assumption 3.3 be satisfied and let the threshold function w and the constant C_α satisfy (29). Further assume that $\hat{\Sigma}_m$ denotes a consistent estimator of the long-run variance $\Sigma_{F^{(1)}}$ based on the observations $X_1, \ldots, X_m$. Under the alternative hypothesis H_1 we have

$$\lim_{m\to\infty} \mathbb{P}\left(\max_{k=0}^{Tm} \frac{\hat{D}_m(k)}{w(k/m)} > C_\alpha\right) = 1.$$

Remark 3.3. We can establish similar results for the statistics (14) and (15) proposed by Wied and Galeano (2013) among others and Fremdt (2014) among others, respectively. For example, if the assumptions of Theorem 3.1 are satisfied we obtain the weak convergence

$$\max_{k=1}^{Tm} \frac{\hat{Q}_m(k)}{w(k/m)} \stackrel{\mathcal{D}}{\Longrightarrow} \sup_{t\in[1,T+1]} \frac{B(t,1)^\top B(t,1)}{w(t-1)},$$

$$\max_{k=1}^{Tm} \frac{\hat{P}_m(k)}{w(k/m)} \stackrel{\mathcal{D}}{\Longrightarrow} \sup_{t\in[1,T+1]} \sup_{s\in[1,t]} \frac{\big(B(1,s) + B(t,1)\big)^\top \big(B(1,s) + B(t,1)\big)}{w(t-1)}$$

under the null hypothesis, which can be used to construct an asymptotic level α monitoring scheme based on the statistics $\hat{Q}_m$ and $\hat{P}_m$, respectively. Consistency of the corresponding tests follows along the arguments given in the proof of Theorem 3.2. The details are omitted for the sake of brevity. The finite sample properties of the three different tests will be investigated by means of a simulation study in Section 5.

3.2. Self-Normalization

The test proposed in Section 3.1 requires an estimator of the long-run variance $\hat{\Sigma}_m$, and we discuss commonly used estimates for this purpose in Section 4. However, it has been pointed out by several authors that this problem is not an easy one as the common estimates depend sensitively on a regularization parameter (e.g., a bandwidth), which might be difficult to select in practice. An alternative to long-run variance estimation is the concept of self-normalization, which will be investigated in this section. This approach has been studied intensively for a-posteriori change point analysis (see Shao and Zhang 2010; Shao

2015, among many others), but—to our best knowledge—self-normalization has not been considered in the context of sequential monitoring. In the following we discuss a self-normalized version of the statistic $\hat{D}_m$ proposed in this article (see Equation (9)). Self-normalization of the statistic $\hat{P}_m$ in (15) will be briefly discussed in Remark 3.4.

To be precise, we define a self-normalizing matrix

$$\mathbb{V}(z,u) = \sum_{j=1}^{z} j^2(z-j)^2\big(\hat{\theta}_1^j - \hat{\theta}_{j+1}^z\big)\big(\hat{\theta}_1^j - \hat{\theta}_{j+1}^z\big)^\top + \sum_{j=z+1}^{u} (u-j)^2(j-z)^2\big(\hat{\theta}_{z+1}^j - \hat{\theta}_{j+1}^u\big)\big(\hat{\theta}_{z+1}^j - \hat{\theta}_{j+1}^u\big)^\top \tag{34}$$

and replace the estimate $\hat{\Sigma}_m$ of the long-run variance in (9) by the matrix $\frac{1}{m^4}\mathbb{V}(m+j,m+k)$. This yields the self-normalized statistic

$$\hat{D}^{\mathrm{SN}}(k) = m \max_{j=0}^{k-1} (m+j)^2(k-j)^2\big(\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\big)^\top \times \mathbb{V}^{-1}(m+j,m+k)\big(\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\big). \tag{35}$$

Theorem 3.3. Let $w : [0,T] \to \mathbb{R}^+$ denote any threshold function, which is increasing and let the assumptions of Theorem 3.1 be satisfied. If the null hypothesis in (2) holds, then as $m \to \infty$

$$\max_{k=1}^{Tm} \frac{\hat{D}_m^{\mathrm{SN}}(k)}{w(k/m)} \overset{\mathcal{D}}{\Longrightarrow} \sup_{t\in[1,T+1]} \sup_{s\in[1,t]} \frac{|\tilde{B}(s,t)|}{w(t-1)}, \tag{36}$$

where

$$\tilde{B}(s,t) = B^\top(s,t)\big(N_1(s) + N_2(s,t)\big)^{-1} B(s,t),$$

the process $\{B(s,t)\}_{(s,t)\in\Delta_2}$ is defined in (26) and $\{N_1(s)\}_{s\in[0,T+1]}$ and $\{N_2(s,t)\}_{(s,t)\in\Delta_2}$ are given by

$$N_1(s) = \int_0^s B(r,s)B^\top(r,s)dr,$$

$$N_2(s,t) = \int_s^t \big(B(r,t) + B(s,r) - B(s,t)\big)\big(B(r,t) + B(s,r) - B(s,t)\big)^\top dr. \tag{37}$$

The monitoring rule is now defined in the same way as described in Section 3.1 determining a constant C_α depending on the threshold function $w(\cdot)$, such that

$$\mathbb{P}\left(\sup_{t\in[1,T+1]} \sup_{s\in[1,t]} \frac{|\tilde{B}(s,t)|}{w(t-1)} > C_\alpha\right) = \alpha \tag{38}$$

for a given level α, and rejecting the null hypothesis H_0 in (2) at the time k, if

$$\hat{D}_m^{\mathrm{SN}}(k) > C_\alpha w(k/m). \tag{39}$$

By Theorem 3.3 this test has asymptotic level α and our next result shows that this procedure is also consistent.

Theorem 3.4. Let Assumption 3.3 be satisfied and let the threshold w and the constant C_α fulfill (38). Under the alternative hypothesis H_1 we have

$$\lim_{m\to\infty} \mathbb{P}\left(\max_{k=0}^{Tm} \frac{\hat{D}_m^{\mathrm{SN}}(k)}{w(k/m)} > C_\alpha\right) = 1.$$

Remark 3.4. The statistic $\hat{P}_m(k)$ defined in (15) can be self-normalized in a similar manner, that is

$$\hat{P}_m^{\mathrm{SN}}(k) = m^3 \max_{j=0}^{k-1} (k-j)^2\big(\hat{\theta}_1^m - \hat{\theta}_{m+j+1}^{m+k}\big)^\top \times \mathbb{V}^{-1}(m+j,m+k)\big(\hat{\theta}_1^m - \hat{\theta}_{m+j+1}^{m+k}\big). \tag{40}$$

If the null hypothesis holds and the assumptions of Theorem 3.1 are satisfied it can be shown using (27) and similar arguments as given in the proof of Theorem 3.3 that

$$\max_{k=1}^{Tm} \frac{\hat{P}_m^{\mathrm{SN}}(k)}{w(k/m)} \overset{\mathcal{D}}{\Longrightarrow} \sup_{t\in[1,T+1]} \sup_{s\in[1,t]} \frac{|\tilde{B}^{\mathrm{SN}}(s,t)|}{w(t-1)}, \tag{41}$$

where the process $\tilde{B}^{\mathrm{SN}}$ is defined by

$$\tilde{B}^{\mathrm{SN}}(s,t) = \big(B(1,s) + B(t,1)\big)^\top \big(N_1(s) + N_2(s,t)\big)^{-1} \big(B(1,s) + B(t,1)\big).$$

Similarly, consistency follows along the lines given in the proof of Theorem 3.4. The details are omitted for the sake of brevity.

On the other hand a statistic like $\hat{Q}_m(k)$ defined in (14) cannot be self-normalized in a straightforward manner as it does not employ a maximum, which is necessary to separate points before and after the (possible) change point. In particular one cannot use the matrix $\mathbb{V}$ in (34), which is based on such a separation. Obviously, a self-normalization approach without separation could be constructed but this would lead to a severe loss in power and is therefore not discussed here. We refer the reader to Shao and Zhang (2010) for a comprehensive discussion of this problem. The finite sample properties of both self-normalized methods $\hat{D}_m^{\mathrm{SN}}(k)$ and $\hat{P}_m^{\mathrm{SN}}(k)$ will be compared by means of a simulation study in Section 5.

3.3. Implementation

We will close this section with a description of the algorithm to detect changes in the functional $\theta(F)$ employing self-normalization. In the following paragraph $\hat{S}_m(k)$ denotes any of the statistics $\hat{D}_m(k), \hat{D}_m^{\mathrm{SN}}(k), \hat{P}_m(k), \hat{P}_m^{\mathrm{SN}}(k)$, and $\hat{Q}_m(k)$ discussed in Sections 3.1 and 3.2.

Algorithm 3.1. Let $\{X_1,\ldots,X_m\}$ denote the "stable" training data satisfying (1).

(*Initialization*) Choose the factor T to determine how much longer the monitoring can be performed. Further choose a threshold function w and a corresponding constant C_α, such that the probability of Type I error is asymptotically α.

(*Monitoring*) If X_{m+k} has been observed, compute the statistic $\hat{S}_m(k)$ and reject the null hypothesis of no change in the parameter $\theta(F)$ if $\hat{S}_m(k) > C_\alpha w(k/m)$. In this case stop monitoring. Otherwise, repeat this comparison with the next observation X_{m+k+1}.

(*Stop*) If there has been no rejection at time $m + mT$, stop monitoring with observation X_{m+mT} and conclude that no change has occurred within the monitoring period.

4. Some Specific Change Point Problems

In this section, we illustrate how the assumptions of Section 3 can be verified for concrete functionals. Exemplarily we consider the mean, variance, and quantiles, but similar arguments could be given for other functionals under consideration. To be precise consider a time series $\{X_t\}_{t\in\mathbb{Z}}$, which forms a physical system in the sense of Wu (2005a), that is

$$X_t = \begin{cases} g(\varepsilon_t, \varepsilon_{t-1}, \ldots) & \text{if } t < \lfloor mc \rfloor, \\ h(\varepsilon_t, \varepsilon_{t-1}, \ldots) & \text{if } t \geq \lfloor mc \rfloor, \end{cases} \tag{42}$$

where $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ denotes a sequence of iid random variables with values in some measure space $\mathbb{S}$ such that the functions $g, h : \mathbb{S}^{\mathbb{N}} \to \mathbb{R}^d$ are measurable. The functions g and h determine the phases of the physical system before and after the change at position $\lfloor mc \rfloor$ with $c > 1$, respectively. Under the null hypothesis we will always assume that g and h coincide, which yields that the (whole) times series $\{X_t\}_{t\in\mathbb{Z}}$ is strictly stationary. In the case $g \neq h$ the random variables X_t form a triangular array, but for the sake of readability we do not reflect this in our notation. To adapt the concept of physical dependence to the situation considered in this article, let ε_0' be an independent copy of ε_0 and define the distances

$$\begin{aligned} \delta_{t,q}^{(1)} &= \left(\mathbb{E}[|g(\varepsilon_t, \varepsilon_{t-1}, \ldots) - g(\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon_1, \varepsilon_0', \varepsilon_{-1}, \ldots)|^q]\right)^{1/q}, \\ \delta_{t,q}^{(2)} &= \left(\mathbb{E}[|h(\varepsilon_t, \varepsilon_{t-1}, \ldots) - h(\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon_1, \varepsilon_0', \varepsilon_{-1}, \ldots)|^q]\right)^{1/q}, \end{aligned} \tag{43}$$

which are used to quantify the (temporal) dependence within both phases of the physical system and $\delta_{t,q}^{(\ell)}$ measures the influence of ε_0 on the random variable X_t. Further let

$$\Theta_q^{(\ell)} = \sum_{t=1}^{\infty} \delta_{t,q}^{(\ell)}, \quad \ell = 1, 2, \tag{44}$$

denote the sum of the coefficients (which might diverge). Additionally, we define the (ordinary) long-run variance matrix of the phases before and after the change by

$$\begin{aligned} \Gamma(g) &= \sum_{t\in\mathbb{Z}} \operatorname{cov}\left(g(\varepsilon_0, \varepsilon_{-1}, \ldots), g(\varepsilon_t, \varepsilon_{t-1}, \ldots)\right), \\ \Gamma(h) &= \sum_{t\in\mathbb{Z}} \operatorname{cov}\left(h(\varepsilon_0, \varepsilon_{-1}, \ldots), h(\varepsilon_t, \varepsilon_{t-1}, \ldots)\right). \end{aligned} \tag{45}$$

4.1. Sequential Testing for Changes in the Mean Vector

In this section, we are interested in detecting changes in the mean

$$\mu(F_t) = \mathbb{E}[X_t] = \int_{\mathbb{R}^d} x dF_t(x), \quad t = 1, 2, \ldots \tag{46}$$

of a d-dimensional time series $\{X_t\}_{t\in\mathbb{Z}}$. Sequential detection schemes for a change in the mean have been investigated by Chu, Stinchcombe, and White (1996), Horváth et al. (2004), and Aue, Horváth, and Reimherr (2009), among others. We consider the closed-end-procedure developed in Section 3 and assume the first m observations $X_1, \ldots, X_m$ to be mean-stable.

As the mean functional (46) is linear the influence function is given by

$$\mathcal{IF}(x, F, \mu) = x - \mu(F) = x - \mathbb{E}_F[X],$$

and therefore Assumption 3.2 is obviously satisfied (note that $R_{i,j} = 0$ for all i, j). Assumption 3.1 reduces to Donsker's invariance principle, that is

$$\left\{\frac{1}{\sqrt{m}} \sum_{t=1}^{\lfloor ms \rfloor} (X_t - \mathbb{E}[X_t])\right\}_{s\in[0,T+1]} \xrightarrow{\mathcal{D}} \left\{\Sigma_F^{1/2} W(s)\right\}_{s\in[0,T+1]} \tag{47}$$

in $\ell^\infty([0, T+1], \mathbb{R}^d)$ with $\Sigma_F = \sum_{t\in\mathbb{Z}} \operatorname{cov}(X_0, X_t)$, which has been derived by Wu (2005a) for physical systems under the assumption that $\Theta_2^{(1)} < \infty$ (see Theorem 3 in this reference). Note that for this functional the (ordinary) long-run variance matrix $\Gamma(g)$ and Σ_F coincide (under stationarity).

If the alternative of a change in the mean at position $\lfloor mc \rfloor$ for some $c \in (1, T+1)$ holds, we may assume that

$$h = g + \Delta\mu, \tag{48}$$

where $\Delta\mu = \mathbb{E}[X_{\lfloor mc \rfloor}] - \mathbb{E}[X_{\lfloor mc \rfloor - 1}]$. Consequently, we have $\Theta_2^{(1)} = \Theta_2^{(2)}$ and if $\Theta_2^{(1)} < \infty$, Assumption 3.3 is also satisfied with $W_1 = W = W_2$ (see also the discussion at the end of Remark 3.2). We summarize these observations in the following proposition.

Proposition 4.1. Assume that (42) holds with $\Theta_2^{(1)} < \infty$ and further let $\hat{\Sigma}_m$ denote a consistent estimator of the (positive definite) long-run variance matrix $\Sigma_{F^{(1)}}$ (before the change) based on the observations $X_1, \ldots, X_m$.

(a) If $g = h$, then the assumptions of Corollary 3.1 and Theorem 3.3 are satisfied for the functional (46). In other words: the sequential tests for a change in the mean based on the statistics $\hat{D}$ or $\hat{D}^{\mathrm{SN}}$ with $\theta(F_t) = \int_{\mathbb{R}^d} x dF_t(x)$ have asymptotic level α.
(b) Let representation (48) hold with $\Delta\mu \neq 0$, then the assumptions of Theorems 3.2 and 3.4 are satisfied for the functional (46). In other words: the sequential tests for a change in the mean based on the statistics $\hat{D}$ or $\hat{D}^{\mathrm{SN}}$ are consistent.

The finite sample properties of this test will be investigated in Section 5.1.

4.2. Sequential Testing for Changes in the Variance

In this section, we focus on detecting changes in the variance. Following Aue et al. (2009), who investigated this problem in the nonsequential case, we consider a time series $\{X_t\}_{t\in\mathbb{Z}}$ with common mean $\mu = \mathbb{E}_F[X]$ and define the functional

$$V(F) = \int_{\mathbb{R}^d} xx^\top dF(x) - \int_{\mathbb{R}^d} x dF(x) \int_{\mathbb{R}^d} x^\top dF(x). \tag{49}$$

A careful but straightforward calculation shows that the corresponding influence function is given by

$$\begin{aligned}\mathcal{IF}(x,F,V) =& -\mathbb{E}_F[XX^\top] + 2\mathbb{E}_F[X]\mathbb{E}_F[X^\top] + xx^\top \\ &- \mathbb{E}_F[X]x^\top - x\mathbb{E}_F[X]^\top \\ =&(x-\mathbb{E}_F[X])(x-\mathbb{E}_F[X])^\top - V(F). \end{aligned} \tag{50}$$

Hence, the remainder term (under stationarity with $X_1 \sim F$) in Equation (19) is given by

$$\begin{aligned} R_{i,j} &= V(\hat{F}_i^j) - V(F) - \frac{1}{j-i+1}\sum_{t=i}^{j}\mathcal{IF}(X_t,F,V) \\ &= \int_{\mathbb{R}^d} xx^\top d\hat{F}_i^j(x) - \int_{\mathbb{R}^d} x d\hat{F}_i^j(x)\int_{\mathbb{R}^d} x^\top d\hat{F}_i^j(x) \\ &\quad - \frac{1}{j-i+1}\sum_{t=i}^{j}(X_i-\mathbb{E}[X_1])(X_i-\mathbb{E}[X_1])^\top \\ &= -\int_{\mathbb{R}^d} x d\hat{F}_i^j(x)\int_{\mathbb{R}^d} x^\top d\hat{F}_i^j(x) + \int_{\mathbb{R}^d} x\mathbb{E}[X_1^\top]d\hat{F}_i^j(x) \\ &\quad + \int_{\mathbb{R}^d}\mathbb{E}[X_1]x^\top d\hat{F}_i^j(x) - \mathbb{E}[X_1]\mathbb{E}[X_1^\top] \\ &= -\left(\int_{\mathbb{R}^d} x - \mathbb{E}[X_1]d\hat{F}_i^j(x)\right)\left(\int_{\mathbb{R}^d} x^\top - \mathbb{E}[X_1^\top]d\hat{F}_i^j(x)\right). \end{aligned} \tag{51}$$

Define vech$(\cdot)$ to be the operator that stacks the columns of a symmetric $d \times d$-matrix above the diagonal as a vector of dimension $d(d+1)/2$. As this operator is linear, it is obvious that expansion (51) is equivalent to

$$\begin{aligned}\mathrm{vech}(R_{i,j}) =& \mathrm{vech}(V(\hat{F}_i^j)) - \mathrm{vech}(V) \\ &- \frac{1}{j-i+1}\sum_{t=i}^{j}\mathcal{IF}_v(X_t,F,V), \end{aligned} \tag{52}$$

where $\mathcal{IF}_v$ is defined as

$$\mathcal{IF}_v(X_t,F,V) = \mathrm{vech}\left(\mathcal{IF}(X_t,F,V)\right) = \mathcal{IF}\big(X_t,F,\mathrm{vech}(V)\big) \tag{53}$$

$$\begin{pmatrix} (X_{t,1}-\mathbb{E}[X_{t,1}])^2 - \mathbb{E}\big[(X_{t,1}-\mathbb{E}[X_{t,1}])^2\big] \\ (X_{t,1}-\mathbb{E}[X_{t,1}])(X_{t,2}-\mathbb{E}[X_{t,2}]) - \mathbb{E}\big[(X_{t,1}-\mathbb{E}[X_{t,1}])(X_{t,2}-\mathbb{E}[X_{t,2}])\big] \\ (X_{t,2}-\mathbb{E}[X_{t,2}])^2 - \mathbb{E}\big[(X_{t,2}-\mathbb{E}[X_{t,2}])^2\big] \\ (X_{t,1}-\mathbb{E}[X_{t,1}])(X_{t,3}-\mathbb{E}[X_{t,3}]) - \mathbb{E}\big[(X_{t,1}-\mathbb{E}[X_{t,1}])(X_{t,3}-\mathbb{E}[X_{t,3}])\big] \\ (X_{t,2}-\mathbb{E}[X_{t,2}])(X_{t,3}-\mathbb{E}[X_{t,3}]) - \mathbb{E}\big[(X_{t,2}-\mathbb{E}[X_{t,2}])(X_{t,3}-\mathbb{E}[X_{t,3}])\big] \\ (X_{t,3}-\mathbb{E}[X_{t,3}])^2 - \mathbb{E}\big[(X_{t,3}-\mathbb{E}[X_{t,3}])^2\big] \\ \vdots \end{pmatrix},$$

and $X_{t,h}$ denotes the hth component of the vector X_t. We now provide sufficient conditions such that the general theory in Section 3 is applicable for the functional vech(V). Assumption 3.1 is satisfied if the time series $\{X_t\}_{t\in\mathbb{Z}}$ is stationary and the invariance principle

$$\frac{1}{\sqrt{m}}\sum_{t=1}^{\lfloor ms\rfloor}\mathcal{IF}_v(X_t,F,V) \xrightarrow{\mathcal{D}} \sqrt{\Sigma_F}W(s), \tag{54}$$

holds in the space $\ell^\infty(\mathbb{R},\mathbb{R}^{d^*})$, where W is a $d^* = d(d+1)/2$-dimensional Brownian motion and Σ_F is defined in (18) with $\mathcal{IF} = \mathcal{IF}_v$. Invariance principles of the form (54) are well known for many classes of weakly dependent time series. The required assumptions for the underlying time series $\{X_t\}_{t\in\mathbb{Z}}$ are typically the same as for the mean—except for some extra moment conditions to cover the product structure of the random variables in (53). Condition (20) in Assumption 3.2 reads as follows

$$\begin{aligned}\sup_{1\le i<j\le n}\frac{1}{j-i+1}&\left|\sum_{t=i}^{j}X_{t,k} - \mathbb{E}[X_{t,k}]\right| \\ &\times\left|\sum_{t=i}^{j}X_{t,\ell} - \mathbb{E}[X_{t,\ell}]\right| = o_{\mathbb{P}}(n^{1/2}) \end{aligned} \tag{55}$$

$(1 \le k,\ell \le d^*)$. The validity of this assumption depends on the underlying dependence structure, in particular of the properties of the functions g and h in (42), and exemplarily we give sufficient conditions in the following result, which is proved in the online supplement.

Proposition 4.2. Assume that (42) holds with bounded functions g and h with $\delta_{t,4}^{(1)} = \mathcal{O}(\rho^t), \delta_{t,4}^{(2)} = \mathcal{O}(\rho^t)$ for some $\rho \in (0,1)$. Let $\hat{\Sigma}_m$ denote a consistent estimator of the long-run variance $\Sigma_{F^{(1)}}$ (before the change) based on the observations $X_1,\ldots,X_m$. Further assume, that the covariance matrices $\Gamma(g)$ and $\Gamma(h)$ defined in (45) are positive definite.

(a) If $g = h$, then the assumptions of Corollary 3.1 and Theorem 3.3 are satisfied for the functional (49). In other words: the sequential tests for a change in the variance based on the statistics $\hat{D}$ or $\hat{D}^{\mathrm{SN}}$ have asymptotic level α.
(b) If $h = A\cdot g$ for some nonsingular matrix $A \in \mathbb{R}^{d\times d}$ with $A\cdot V(F^{(1)})\cdot A^\top \neq V(F^{(1)})$, then the assumptions of Theorems 3.2 and 3.4 are satisfied for the variance functional (49). In other words: the sequential tests for a change in the variance based on the statistics $\hat{D}$ or $\hat{D}^{\mathrm{SN}}$ are consistent.

Remark 4.1. The assumption of bounded observations in Proposition 4.2 is crucial to prove the estimate (55). Essentially a proof of such a statement requires a version of Theorem 1 in Shao (1995) for dependent random variables. The main ingredient for a proof of Shao's result is an Erdös–Renyi–Law of large numbers in the case of dependent random variables, which—to the authors best knowledge—is only known for bounded random variables (see, e.g., Kifer 2017). On the other hand the assumption of bounded functions g and h in (42) is not necessary for the functional (49) in the case of M-dependent time series.

4.3. Sequential Testing for Changes in the Quantiles

In this section, we consider the quantile functional

$$\varphi_\beta(F) = F^-(\beta) := \inf\{x\in\mathbb{R} \mid F(x) \ge \beta\}, \tag{56}$$

where $\beta \in (0,1)$ is fixed and F^- denotes the quantile function (or sometimes called generalized inverse function) for a distribution function F. Further recall the notation $\hat{F}_i^j$ defined in (10) for the empirical distribution function based on the subset of observations $X_i,\ldots,X_j$. For the sake of readability, we

will simplify the notation and denote the true and empirical quantiles by

$$q_\beta := \varphi_\beta(F) = F^-(\beta) \quad \text{and} \quad [\hat{q_\beta}]_i^j := \varphi_\beta(\hat{F}_i^j) = (\hat{F}_i^j)^-(\beta),$$

respectively. Considering a twice differentiable distribution function F having derivative f with $f(q_\beta) > 0$, a straightforward but tedious calculation yields that the influence functional for φ_β is given by

$$\mathcal{IF}(x, F, \varphi_\beta) = \frac{\beta - I\{x \le q_\beta\}}{f(q_\beta)} = \begin{cases} \dfrac{\beta - 1}{f(q_\beta)} & \text{if } x \le q_\beta, \\ \dfrac{\beta}{f(q_\beta)} & \text{if } x > q_\beta \end{cases}$$

(see, e.g., Wasserman 2010 for a proof of this statement under slightly stronger conditions). This yields that the linearization in (19) for φ_β is given by

$$\begin{aligned} \varphi_\beta(\hat{F}_i^j) - \varphi_\beta(F) &= [\hat{q_\beta}]_i^j - q_\beta \\ &= \frac{1}{j-i+1} \sum_{t=i}^{j} \mathcal{IF}(X_t, F, \varphi_\beta) - R_{i,j} \\ &= \frac{\beta - \hat{F}_i^j(q_\beta)}{f(q_\beta)} - R_{i,j}, \end{aligned}$$

with remainder terms

$$R_{i,j} = q_\beta - [\hat{q_\beta}]_i^j - \frac{\beta - \hat{F}_i^j(q_\beta)}{f(q_\beta)}.$$

The linearization stated above is also known as Bahadur expansion (see Bahadur 1966) and the investigation of the order of the remainder terms has been a major research topic. In this section, we restrict ourselves to the case of independent observations, which makes the arguments less technical (see the discussion in the online supplement). Given independent and identically distributed observations, an application of Donsker's theorem immediately shows that Assumption 3.1 is satisfied for φ_β, where it suffices to use the bound

$$|\mathcal{IF}(X_t, F, \varphi_\beta)| = \left| \frac{\beta - I\{X_t \le q_\beta\}}{f(q_\beta)} \right| \le 2/f(q_\beta).$$

Establishing Assumption 3.2 is substantially more complicated. To the author's best knowledge uniform estimates for the remainder terms of the Bahadur expansion of the form (20) have not been investigated in the literature. In the online supplement we prove the following result, which is of independent interest.

Theorem 4.1. Let $\{X_t\}_{t\in\mathbb{Z}}$ be a sequence of iid random variables with distribution function F, which is twice differentiable with

$$\sup_{x\in\mathbb{R}} \big(f(x) + |f'(x)|\big) < \infty \quad \text{and} \quad \mathbb{P}\big(|X_1| > x\big) \lesssim x^{-\lambda} \tag{57}$$

for fixed $\lambda > 18/5$. Further let $\beta \in (0,1)$ be fixed and assume that $f(q_\beta) > 0$. It holds that

$$\frac{1}{\sqrt{n}} \max_{1\le i<j\le n} (j-i+1)|R_{i,j}| = \frac{1}{\sqrt{n}} \max_{1\le i<j\le n} (j-i+1) \left| [\hat{q_\beta}]_i^j - q_\beta - \frac{\beta - \hat{F}_i^j(q_\beta)}{f(q_\beta)} \right| = o_{\mathbb{P}}(1).$$

The assumption on the tails of the distribution of $|X_1|$ is crucial to control the error of the quantile estimators in case of small sample sizes, see Lemma A.7 in the online supplement. The theorem above directly implies the following corollary.

Corollary 4.1. Let $\{X_t\}_{t\in\mathbb{Z}}$ be a sequence of independent random variables and $F^{(1)}, F^{(2)}$ distributions functions fulfilling the assumptions of Theorem 4.1. Assume that for a constant $1 < c < T$

$$X_t \sim \begin{cases} F^{(1)} & \text{if } t < \lfloor mc \rfloor, \\ F^{(2)} & \text{if } t \ge \lfloor mc \rfloor. \end{cases} \tag{58}$$

Further let $\hat{\Sigma}_m$ denote a consistent estimator of the long-run variance $\Sigma_{F^{(1)}}$ (before a possible change) based on the observations $X_1, \ldots, X_m$.

(a) If $F^{(1)} = F^{(2)}$, then the assumptions of Corollary 3.1 and Theorem 3.3 are satisfied for the functional (56). In other words: the sequential tests for a change in the β-quantile based on the statistics $\hat{D}$ or $\hat{D}^{\mathrm{SN}}$ have asymptotic level α.
(b) If $\varphi_\beta(F^{(1)}) \neq \varphi_\beta(F^{(2)})$, then the assumptions of Theorems 3.2 and 3.4 are satisfied for the functional (56). In other words: the sequential tests for a change in the β-quantile based on the statistics $\hat{D}$ or $\hat{D}^{\mathrm{SN}}$ are consistent.

5. Finite Sample Properties

In this section, we investigate the finite sample properties of the new detection schemes based on the statistics $\hat{D}$ and $\hat{D}^{\mathrm{SN}}$ in (9) and (35) and also provide a comparison to the detection schemes based on $\hat{Q}$, $\hat{P}$, and $\hat{P}^{\mathrm{SN}}$, which are defined in (14), (15), and (40), respectively. For the choice of the threshold function we follow the ideas of Horváth et al. (2004), Aue, Horváth, and Reimherr (2009), and Wied and Galeano (2013) and consider the parametric family

$$w(t) = (t+1)^2 \cdot \max\left\{ \left(\frac{t}{t+1}\right)^{2\gamma}, \delta \right\}, \tag{59}$$

where the parameter γ varies in the interval $[0, 1/2)$ and $\delta > 0$ is a small constant introduced to avoid problems in the denominator of the ratio considered in (29). For the statistic $\hat{Q}$ these threshold functions are motivated by the law of iterated logarithm and are used to reduce the stopping delay under the alternative hypothesis (see Aue, Horváth, and Reimherr 2009 or Wied and Galeano 2013).

Note that we use the squared versions of the thresholds from the cited references, since we consider statistics in terms of quadratic forms. To be precise consider three different threshold functions

(T1) $w(t) = 1$,
(T2) $w(t) = (t+1)^2$,
(T3) $w(t) = (t+1)^2 \cdot \max\left\{ \left(\frac{t}{t+1}\right)^{1/2}, 10^{-10} \right\}$,

and the constant C_α, which determines the rejection region, is chosen by Monte Carlo simulations, such that

$$\mathbb{P}\left(\sup_{t\in[1,T+1]} \frac{L(t)}{w(t-1)} > C_\alpha \right) = \alpha \tag{60}$$

with L denoting the limit process corresponding to the test statistic under consideration. For the estimation of the long-run variance of a one-dimensional time series we use the well-known quadratic spectral kernel (see Andrews 1991), that is

$$\hat{\sigma}^2 = \hat{\gamma}_0 + 2\sum_{i=1}^{m-1} k\Big(\frac{i}{b_m}\Big)\hat{\gamma}_i, \tag{61}$$

where $\hat{\gamma}_i$ denotes the empirical lag i autocovariance of $X_1, \ldots, X_m$ and b_m is the bandwidth for the underlying kernel k given by

$$k(x) = \frac{25}{12\pi^2 x^2}\left(\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5)\right).$$

Note, that we only use the data from the stable subset $X_1, \ldots, X_m$ for the long-run variance estimate, which is common practice in sequential monitoring (see Aue et al. 2006; Wied and Galeano 2013; Fremdt 2014; Pape, Wied, and Galeano 2016, among many others). For multivariate data, we use its canonical extension replacing the estimated autocovariances $\hat{\gamma}$ by its corresponding multivariate counterparts. In our simulations, we use the implementation of this estimator contained in the R-package "sandwich" (see Zeileis 2004). Note that the long-run variance estimator is the same for the non-self-normalized procedures $\hat{D}, \hat{P}, \hat{Q}$. For the sake of brevity, we will only display situations where the parameter T is fixed as $T = 1$, that is, the monitoring period will always have the same size as the historical dataset. Further the change will always occur at the center of the monitoring period, which is $k^* = m/2$. For the case $T \geq 2$ and other change locations we obtained a similar picture. The results can be found in Section B of the online supplement. All results that are presented here and in the online supplement are based on 5000 independent simulation runs.

5.1. Changes in the Mean

For the analysis of the new procedures in the problem of detecting changes in the mean we look at independent data, a MA(2)- and two AR(1)-processes, defined by

(M1) $X_t = \varepsilon_t$,
(M2) $X_t = 0.3X_{t-1} + \varepsilon_t$,
(M3) $X_t = \varepsilon_t + 0.3\varepsilon_{t-1} - 0.1\varepsilon_{t-2}$,
(M4) $X_t = 0.7X_{t-1} + e_t$,

where $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ and $\{e_t\}_{t\in\mathbb{Z}}$ are sequences of independent standard Gaussian and exp(1) distributed random variables, respectively. In the case of the alternative hypothesis we consider the sequence

$$X_t^\mu = \begin{cases} X_t & \text{if } t < m + \lfloor \frac{m}{2} \rfloor \\ X_t + \mu & \text{if } t \geq m + \lfloor \frac{m}{2} \rfloor \end{cases}$$

for various values of μ. If not mentioned otherwise, the bandwidth is chosen as $b_m = \log_{10}(m)$ (this corresponds to the rule proposed in Aue et al. (2009)). For models (M2) and (M4) defined below, we use $b_m = \log_{10}(m^3)$ and $b_m = \log_{10}(m^4)$, respectively, to take into account the stronger dependence of the observations.

For all discussed detection schemes the empirical rejection probabilities for the models (M1)–(M4) and threshold functions (T1)–(T3) are shown in Figures 1 and 2 corresponding to the choice $m = 50$ and $m = 100$ as initial sample size. The results can be summarized as follows. The statistic $\hat{D}$ outperforms $\hat{P}$ and $\hat{Q}$ with respect to the power for all combinations of the model and threshold function. Further the statistic $\hat{P}$ shows a better performance with respect to power as $\hat{Q}$ in all cases under consideration. For example, the plot in the left-upper corner of Figure 1 shows, that $\hat{D}$ already has empirical power close to 1 (0.95) for a change of size 1, while $\hat{P}$ and $\hat{Q}$ have only empirical power of 0.84 and 0.71, respectively. This relation is basically the same in all plots contained in Figures 1 and 2. Note that one can observe a loss of power in model (M4) (see the last row of both figures), which is a consequence of a greater (long-run) variance caused by the stronger dependence structure. As a consequence it is harder to separate structural changes in the mean from random noise.

In Figure 3, we present the average rejection time of the different procedures as a function of the size of the change for the different choices (T1)–(T4) of the threshold function under model (M1). For its computation we ignore runs without a rejection or with a rejection before the actual change. The results basically show that the rejections occur earlier when using threshold (T3) or (T2) instead of the constant threshold (T1). The threshold (T3) also yields slightly earlier rejections compared to threshold (T2). This effect may be caused by a value of γ in the parametric family in (59) ($\gamma = 0$ and $\gamma = 0.25$), which corresponds to the observations made by Horváth et al. (2004) or Wied and Galeano (2013), where a more detailed discussion is given. Basically, the plots illustrate a decreasing rejection with an increasing size of the change provided that this is larger than 0.5. This corresponds to intuition. The slight decrease in the empirical time of rejection for small values of μ can be explained by the fact that this case is close to the null hypotheses. As a consequence the rejection times are more uniformly distributed, with a greater portion close to the time $m + k^*$.

For the sake of brevity, we will only consider the constant threshold (T1) in the remaining part of this article (note also that the results in Figures 1 and 2 show no substantial differences between the different threshold functions). In Figure 4, we display the power of the tests based on the self-normalized statistics $\hat{D}^{\mathrm{SN}}$ and $\hat{P}^{\mathrm{SN}}$ and non-self-normalized statistics $\hat{D}$ and $\hat{P}$. The results are similar as before and the empirical power obtained by the use of $\hat{D}^{\mathrm{SN}}$ is considerably higher than those of $\hat{P}^{\mathrm{SN}}$. Further a comparison between the results of $\hat{D}^{\mathrm{SN}}$ and $\hat{P}^{\mathrm{SN}}$ to those of $\hat{D}$ and $\hat{P}$ indicates, that self-normalization yields a substantial loss of power in the sequential detection schemes.

On the other hand, the approximation of the nominal level is more stable with respect to different dependence structures for self-normalized methods. To illustrate this fact we display in Table 1 the Type I error for all five sequential monitoring schemes based on the statistics $\hat{D}$, $\hat{P}$, and $\hat{Q}$. The results provide some empirical evidence that the self-normalized statistics yield a more stable approximation of the nominal level. In particular for models (M2) and (M4), which have a stronger dependence structure, the self-normalized methods yield a sub-

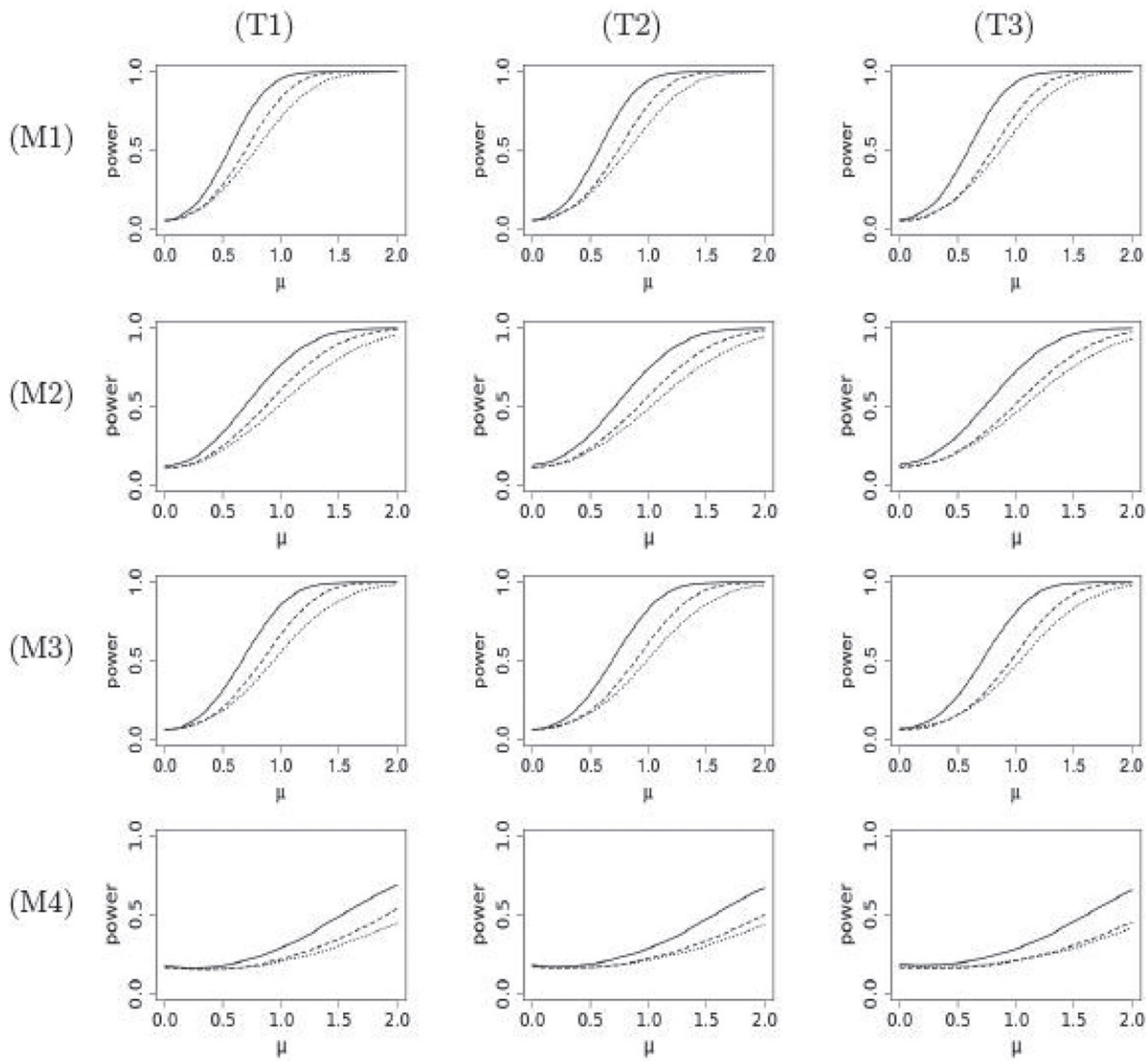

Figure 1. Empirical rejection probabilities of the sequential tests for a change in the mean based on the statistics $\hat{D}$ (solid line), $\hat{P}$ (dashed line), $\hat{Q}$ (dotted line). The initial and total sample size are $m = 50$ and $m(T+1) = 100$, respectively, and the change occurs at observation 75. The level is $\alpha = 0.05$. Different rows correspond to different models, while different columns correspond to different threshold functions.

stantially more precise approximation of the nominal level. This superiority is nearly nonvisible, if smaller AR(1)-coefficients are used in models (M2) and (M4) (these simulation results are not reported for the sake of brevity). We suppose that the approximation of the nominal level by the non-self-normalized statistics $\hat{D}$, $\hat{P}$, and $\hat{Q}$ can be improved by using more refined estimates of the long-run variance, because we observe a very precise approximation of the nominal level if we use the true long-run variance in the simulation. For example, in model (M2) with $m = 100$ (corresponding to row 3 of Table 1) and true long-run variance the simulated Type I error is 3.3%, 4.2%, and 4.7% for the statistics $\hat{D}$, $\hat{P}$, and $\hat{Q}$, respectively.

As pointed out by a reviewer, the method $\hat{D}$ and the self-normalized methods $\hat{D}^{\mathrm{SN}}$, $\hat{P}^{\mathrm{SN}}$ should require higher computational effort. To offer a comparison of the computational complexity, we illustrate the run time of the different procedures in Table 2, where we display the (average) CPU-time of each procedure for a single run. We observe that the computation of the statistic $\hat{D}$ is more expensive than those of $\hat{P}$ since it requires updating $\hat{\theta}_1^{m+j}$ and $\hat{\theta}_{m+j+1}^{m+k}$ for all possible change positions $j \in \{0, \ldots, k-1\}$ when a new observation X_{m+k+1} arrives. In contrast to this, $\hat{P}$ only requires renewing $\hat{\theta}_{m+j+1}^{m+k}$ (for all possible choices of j) while the historical estimate $\hat{\theta}_1^{m}$ is fixed. Note that the procedure based on $\hat{Q}$ is the fastest procedure since there is no maximum included and only the estimation $\hat{\theta}_m^{m+k}$ has to be updated when moving to the next observation X_{m+k+1}. However, one has to keep in mind that the advantage of $\hat{Q}$ and $\hat{P}$ over $\hat{D}$ with respect to computation time comes with the price of power inferiority described above. As expected, the computation time for the self-normalized procedures $\hat{P}$ and $\hat{D}$ is larger, as the process $\mathbb{V}$ defined in (34) is of a more complicated structure.

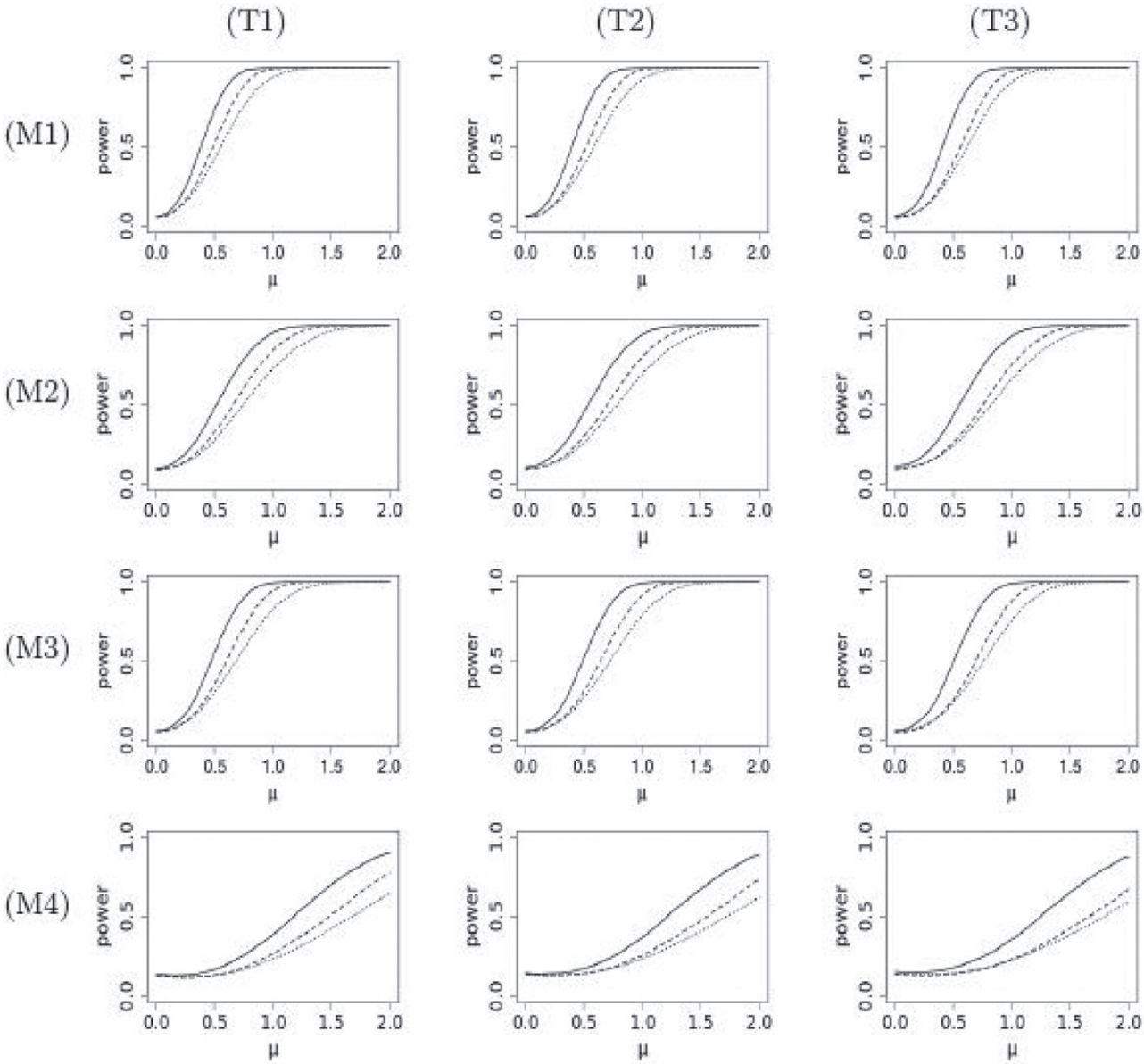

Figure 2. Empirical rejection probabilities of the sequential tests for a change in the mean based on the statistics $\hat{D}$ (solid line), $\hat{P}$ (dashed line), $\hat{Q}$ (dotted line). The initial and total sample size are $m = 100$ and $m(T+1) = 200$, respectively, and the change occurs at observation 150. The level is $\alpha = 0.05$. Different rows correspond to different models, while different columns correspond to different threshold functions.

However, all procedures are quite fast, such that the observed time disparities might have an impact in a large simulation study but will not be of importance in practice when a dataset has to be evaluated.

5.2. Changes in the Variance

In this subsection, we present a small simulation study investigating the performance of the detection schemes for a change in the variance matrix. We consider the following models

(V1) $X_t = \varepsilon_t$,
(V2) $X_t = A_1 X_{t-1} + \varepsilon_t$,
(V3) $X_t = \varepsilon_t + A_2 \varepsilon_{t-1} + A_3 \varepsilon_{t-2}$,
(V4) $X_t = A_4 X_{t-1} + \varepsilon_t$,

where $\{\varepsilon_t\}_{t\in\mathbb{Z}} = \{(\varepsilon_{t,1}, \ldots, \varepsilon_{t,d})^\top\}_{t\in\mathbb{Z}}$ denotes an iid sequence of centered d-dimensional Gaussian distributed random variables and d is chosen accordingly to the dimension of the involved matrices, which are defined by

$$A_1 = \begin{pmatrix} 0.2 & 0.1 \\ 0.1 & 0.2 \end{pmatrix}, \; A_2 = \begin{pmatrix} 0.3 & 0.1 \\ 0.1 & 0.3 \end{pmatrix}, \; A_3 = \begin{pmatrix} 0.1 & 0.05 \\ 0.05 & 0.1 \end{pmatrix},$$

$$A_4 = \begin{pmatrix} 0.1 & 0.05 & 0.05 \\ 0.05 & 0.1 & 0.05 \\ 0.05 & 0.05 & 0.1 \end{pmatrix}. \tag{62}$$

For the alternative, we proceed similarly as in Aue et al. (2009) and define

$$\mathrm{cov}(\varepsilon_t, \varepsilon_t) = \varepsilon_t \cdot \varepsilon_t^\top = \begin{cases} I_d & \text{if } t \le m + \lfloor \frac{m}{2} \rfloor \\ I_d + \delta \cdot I_d & \text{if } t > m + \lfloor \frac{m}{2} \rfloor \end{cases}, \tag{63}$$

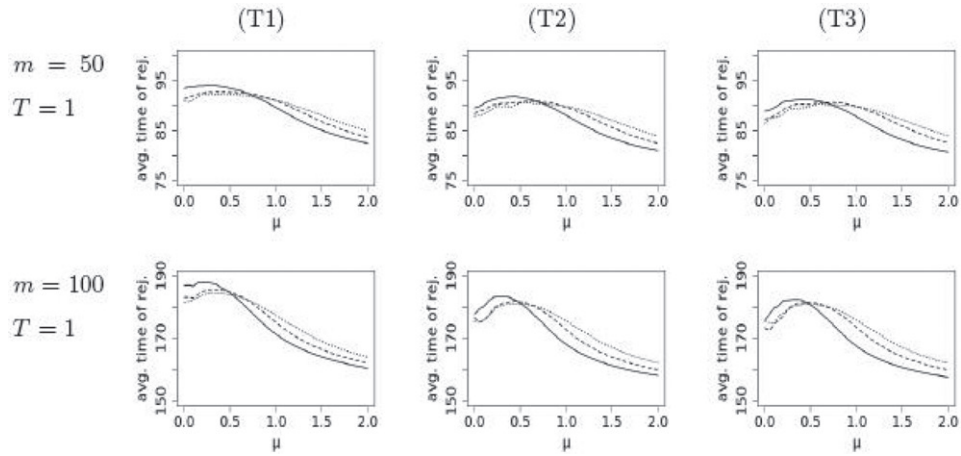

Figure 3. Empirical time of rejection of the sequential tests for a change in the mean based on the statistics $\hat{D}$ (solid line), $\hat{P}$ (dashed line), $\hat{Q}$ (dotted line). The model is chosen as (M1). The change occurs at observation 75 (first row) and 150 (second row), respectively. The level is $\alpha = 0.05$. Different columns correspond to different threshold functions. Simulation runs without a rejection or with a rejection before the actual change are ignored.

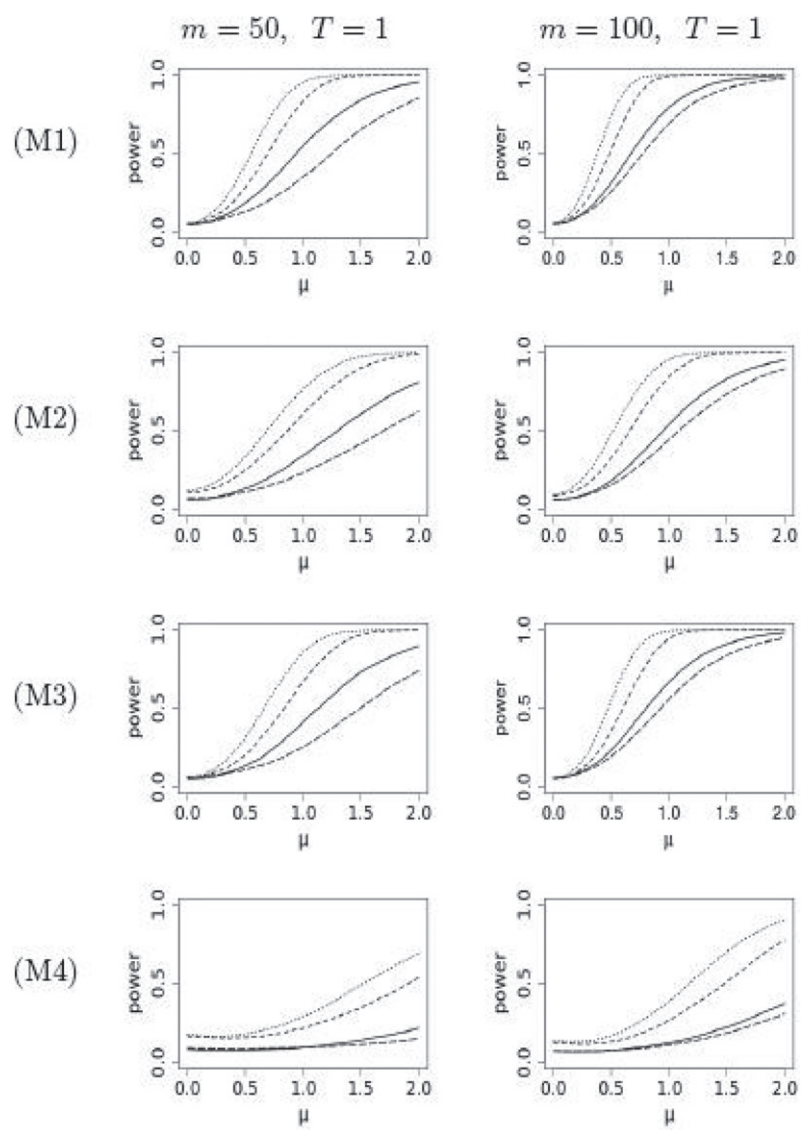

Figure 4. Empirical rejection probabilities of the sequential tests for a change in the mean based on the self-normalized statistics $\hat{D}^{SN}$ (solid line), $\hat{P}^{SN}$ (long dashed line) compared to the non-self-normalized statistics $\hat{D}$ (dotted line) and $\hat{P}$ (dashed lined). The initial and total sample size are $m = 50$ and $m(T+1) = 100$ (upper panel, change at observation 75) and $m = 100$ and $m(T+1) = 200$ (lower panel, change at observation 150). The level is $\alpha = 0.05$ and the threshold function is given by (T1).

Table 1. Simulated Type I error (level $\alpha = 0.05$) of the sequential tests for a change in the mean based on the statistics $\hat{D}, \hat{P}, \hat{Q}, \hat{D}^{SN}$, and $\hat{P}^{SN}$.

		Statistic				
m	Model	$\hat{D}$	$\hat{P}$	$\hat{Q}$	$\hat{D}^{SN}$	$\hat{P}^{SN}$
	(M1)	5.6%	5.3%	5.8%	5.0%	5.7%
50	(M2)	12.2%	10.6%	10.6%	6.0%	7.1%
	(M3)	6.4%	6.3%	6.8%	5.2%	6.3%
	(M4)	18.0%	16.9%	17.2%	8.4%	9.1%
	(M1)	5.9%	5.8%	5.9%	5.1%	5.7%
100	(M2)	9.8%	8.7%	9.0%	6.3%	6.7%
	(M3)	5.8%	5.4%	5.7%	5.9%	6.1%
	(M4)	14.0%	12.5%	12.7%	7.3%	7.5%

NOTE: The threshold function is (T1) and the factor T is again set to $T = 1$.

Table 2. Average computation time for one simulation run of the procedures $\hat{D}, \hat{P}, \hat{Q}, \hat{D}^{SN}, \hat{P}^{SN}$.

$\hat{D}$	0.0119 sec	$\hat{D}^{SN}$	0.3358 sec
$\hat{P}$	0.0090 sec	$\hat{P}^{SN}$	0.3424 sec
$\hat{Q}$	0.0071 sec		

NOTE: Basis is the scenario in the left upper corner of Figure 4.

where I_d denotes the d-dimensional identity matrix (the case $\delta = 0$ corresponds to the null hypothesis of no change). For the sake of brevity, we will focus on the non-self-normalized statistics $\hat{D}$, $\hat{P}$, and $\hat{Q}$ here and use for all models defined above a bandwidth of $b_m = \log_{10}(m)$ for the long-run variance estimate in these statistics. In Figure 5, we display the empirical power for the three data generating processes and the threshold function (T1). The results are similar to those presented in Section 5.1. The test based on the statistic $\hat{D}$ is more powerful than the tests based on $\hat{P}$ and $\hat{Q}$. It should be mentioned that the approximation of the nominal level is less accurate for model (V4).

5.3. Changes in the Correlation

We conclude this article with a brief empirical comparison of the three methods for the detection of a change in the correlation, which has been considered in Wied and Galeano (2013). For the sake of brevity, we do not provide a detailed proof that the assumptions of Section 3 are satisfied, but restrict ourselves to the numerical comparison. For the definition of the data generating processes, we use the models (V1)–(V3) introduced in Section 5.2 but with a different process $\{\varepsilon_j\}_{j\in\mathbb{Z}} = \{(\varepsilon_{j,1}, \varepsilon_{j,2})^\top\}_{j\in\mathbb{Z}}$. In this section, $\{\varepsilon_j\}_{j\in\mathbb{Z}}$ is a sequence of independent two-dimensional Gaussian random variables such that

$$\operatorname{cor}(\varepsilon_{j,1}, \varepsilon_{j,2}) = \begin{cases} c_1 & \text{if} \quad j \le m + \lfloor \frac{m}{2} \rfloor, \\ c_2 & \text{if} \quad j > m + \lfloor \frac{m}{2} \rfloor, \end{cases}$$

and $\operatorname{var}(\epsilon_{j,1}) = \operatorname{var}(\epsilon_{j,2}) = 1$. We use $c_1 = 0.3$ for the correlation before the change and consider different values of c_2. For estimation of the long-run variance matrix, we use the estimator proposed in Wied and Galeano (2013) (the explicit formula for the estimator is given in the appendix of the referenced article

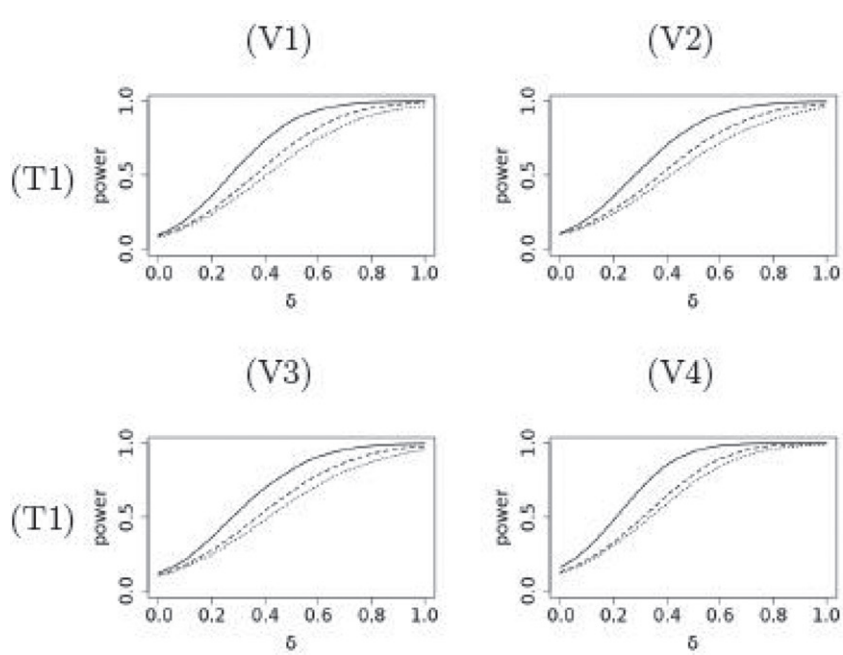

Figure 5. Empirical rejection probabilities of the sequential tests for a change in the variance matrix based on the statistics $\hat{D}$ (solid line), $\hat{P}$ (dashed line), $\hat{Q}$ (dotted line). The initial and total sample size are $m = 200$ and $m(T + 1) = 400$, respectively, and the change occurs at observation 300. The level is $\alpha = 0.05$.

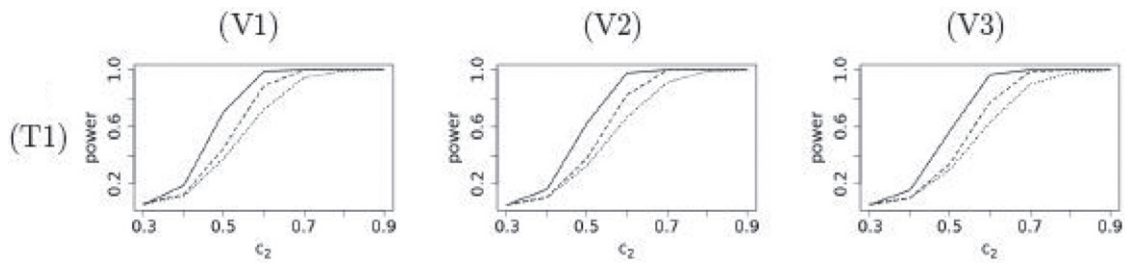

Figure 6. Empirical rejection probabilities of the sequential tests for a change in the correlation based on the statistics $\hat{D}$ (solid line), $\hat{P}$ (dashed line) , $\hat{Q}$ (dotted line). The initial and total sample size are $m = 500$ and $n = 1000$, respectively, and the change occurs at observation 750. The level is $\alpha = 0.05$.

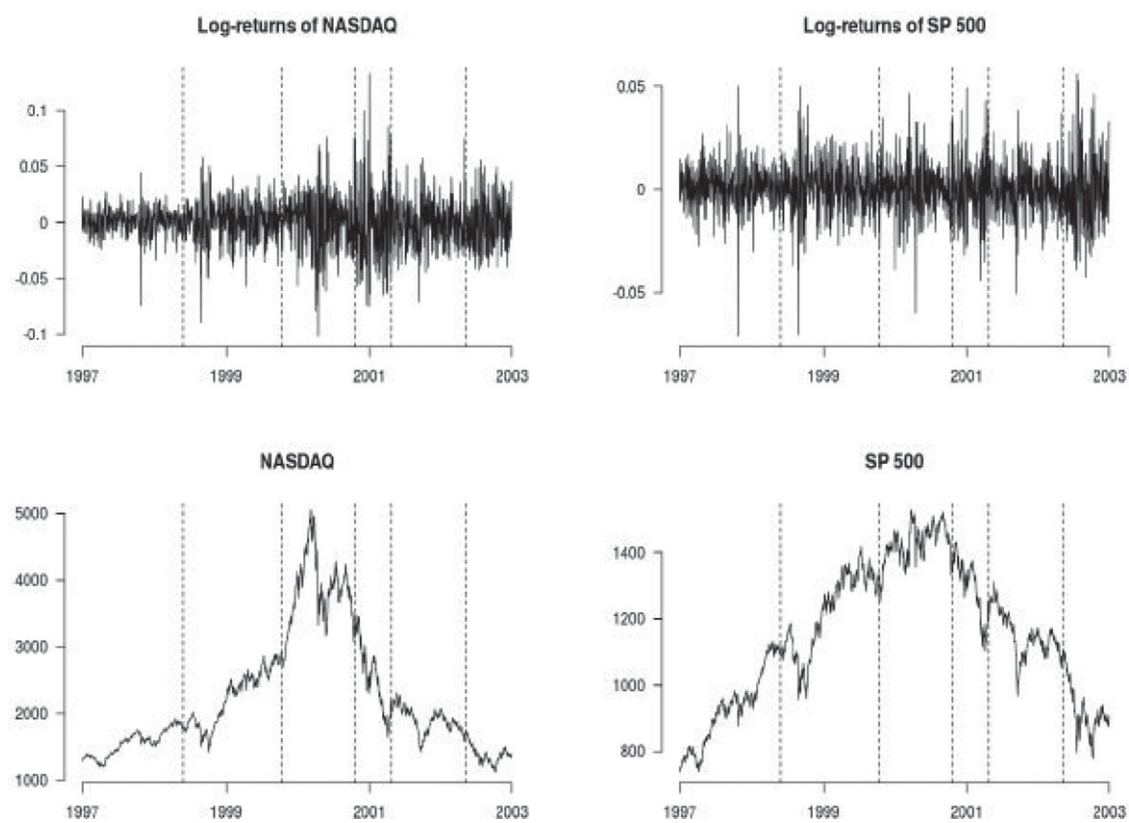

Figure 7. Log-returns (upper row) and prices (lower row) of NASDAQ Composite and Standard and Poor's 500 indices. The vertical dashed lines mark positions of estimated change points found by our analysis with threshold function (T3).

and omitted here for the sake of brevity). Figure 6 now compares the power of the non-self-normalized methods for the three models defined above. As in the previous sections the sequential detection scheme based on $\hat{D}$ yields substantially better results.

5.4. Data Example

In this section, we provide a small data example to illustrate potential applications of the new method based on the statistic $\hat{D}$. For this purpose we consider the log-returns of the NASDAQ Composite Index and the Standard & Poor's 500 Index in the period from 1997-01-02 to 2002-12-31 (rise and burst of dot-com bubble) and investigate this data for potential changes in the covariance matrix. The log-returns and corresponding prices are shown in Figure 7. All data used was obtained from *finance.yahoo.com* on August 11, 2018.

If the monitoring rejects the null hypothesis of a stable covariance matrix (at time k), we directly obtain an estimate of the change point location via the formula

$$\hat{\ell} = m + \operatorname*{argmax}_{j=0}^{k-1}(m+j)^2(k-j)^2\big(\hat{V}_1^{m+j} - \hat{V}_{m+j+1}^{m+k}\big)^\top \times \hat{\Sigma}_m^{-1}\big(\hat{V}_1^{m+j} - \hat{V}_{m+j+1}^{m+k}\big), \tag{64}$$

where $\hat{V}_i^j := \operatorname{vech}(V(\hat{F}_i^j))$ is the vectorized covariance functional (see Equation (49)). Note that (64) can be directly derived from formula (9) with the constant threshold function. It is a canonical modification of the classical argmax estimator (see Jandhyala et al. 2013) from the data $X_1, \ldots, X_{m+k}$, which addressed the fact that there is no change point in the training sample. To verify stability of the historical/training dataset we employ the retrospective test based on the statistic $\tilde{\Lambda}_n$ defined on page 6 in Aue et al. (2009) with the Newey-West estimator for the log-run variance using the automatic bandwidth selection contained in the R-package "sandwich" (see Zeileis 2004).

To be precise, the methodology at hand is applied as follows: If the test of Aue et al. (2009) rejects for the potential initial

Table 3. Rejection dates (found at) and corresponding estimates of the change point locations for the (log-return) covariance matrix of NASDAQ Composite and Standard and Poor's 500 indices in the time frame 1997-01-02 to 2002-12-31 for different choices of threshold functions.

(T1)		(T2)		(T3)	
Change point	Found at	Change point	Found at	Change point	Found at
1998-07-29 (S)	1999-02-12	1998-07-29 (S)	1998-10-01	1998-05-29 (S)	1998-09-08
1999-12-31 (S)	2000-04-03	1999-10-14 (S)	2000-03-15	1999-10-14 (S)	2000-03-14
2000-05-03 (R)	2001-01-04	2001-04-23 (S)	2002-01-31	2000-10-18 (S)	2001-01-03
2000-10-11 (R)	2001-05-08	2002-05-15 (S)	2002-07-22	2001-04-23 (R)	2001-10-29
2001-04-23 (R)	2001-10-22			2002-05-14 (S)	2002-07-19
2002-06-14 (S)	2002-07-10				

NOTE: In brackets we denote if a change point was detected by the sequential (S) or by the retrospective method (R).

stable sample of m observations (where $m = 255$ corresponds approximately to one year), we use the corresponding estimator, say $\hat{\ell}_A$, from this reference to estimate the change point and consider (next) the set $\{X_{\hat{\ell}_A+1}, \ldots, X_{\hat{\ell}_A+m}\}$ to investigate stability. Otherwise, we start the monitoring procedure proposed in this article with the observations $\{X_1, \ldots, X_m\}$ as the stable initial dataset. If this monitoring does not reject, we would simply stop monitoring after $m + mT$ observations (2002-12-31) and conclude that there was no change in the considered time frame. If sequential monitoring rejects, we report the day of rejection and the corresponding location estimate $\hat{\ell}$ defined in (64). Next we define the m ($\approx$ one year) data points $\{X_{\hat{\ell}+1}, \ldots, X_{\hat{\ell}+m}\}$ subsequent of the location estimate $\hat{\ell}$ as a new (potential) historic dataset and restart with the retrospective analysis as described above. The level of all tests is 5%. The results of this approach are listed in Table 3 and the estimated change points are displayed in Figure 7.

For example, the first column in Table 3 is obtained as follows. We start by applying the retrospective test to the first 255 observations (1997-01-02 to 1997-12-31). Since this test does not reject the null hypothesis of a stable covariance matrix, we use this data as first initial sample and start our monitoring procedure with endpoint 2002-12-31 ($m = 255$, $T = 4.92$). This sequential monitoring then rejects with observation 534 (1999-02-12) and the corresponding location estimate for the change is 397 (1998-07-29) (see also Table 3). Next we check (covariance) stability of the observations 398 to 652($= 398 + 254$) with the retrospective test. This test does not reject, which means that this set will be the new historic period. Now the sequential procedure is relaunched with the same endpoint (2002-12-31) but an adapted $T = 3.36$. This procedure is continued until the end of the period under investigation.

6. Conclusion and Outlook

In this article, we have proposed a new closed-end sequential monitoring procedure for changes in the parameter of a d-dimensional time series. We have proved that for large sample sizes the new tests keep its prespecified level and are consistent. Moreover, we introduce the concept of self-normalization for sequential change point detection to avoid the estimation of the long-run variance. Our approach is motivated by the concept of likelihood ratio tests and is generally applicable, whenever the estimate of the parameter can be represented as a linear function of the empirical distribution function with a remainder satisfying several regularity conditions. These assumptions have to be verified for each case individually and we do this here for the mean, variance, and quantile. For the latter, only the iid case is considered and so the natural next step here we would be to extend the results to a setup, where we also allow for dependence. For this purpose it may be possible to adapt more parts of Wu (2005b), who develops a Bahadur representation for dependent observations, which holds uniform in quantile level. An interesting direction for future research is the investigation if these conditions are also satisfied for other functionals such as skewness, kurtosis, or correlation (for the latter we give some numerical results).

Our empirical findings provide strong evidence that compared to the currently available methodology the new sequential detection scheme based on the likelihood ratio approach yields to a substantial improvement with respect to power. The improvement was observed in all examples under consideration. It is less visible if the change points occur shortly after the initial stable sequence but substantial for all other cases, in particular, if the change point is close to the end of the monitoring period. Thus—although the new approach is computationally more expensive—it should be preferred to the currently available methodology.

An important direction for future research is the development of a corresponding methodology for open-end procedures as considered by Chu, Stinchcombe, and White (1996), Horváth et al. (2004), or Kirch and Weber (2018), for example. A further interesting problem is if sequential monitoring schemes could be improved using estimates of the long-run variance from the full sample instead of the training sample. Moreover, it remains to find suitable threshold functions for the monitoring procedure based on $\hat{D}$. The functions considered in this work have been originally developed for other monitoring procedures (see Horváth et al. 2004; Wied and Galeano 2013, among others), and it is likely that the performance of the new procedure can be further improved by choosing an alternative weight function.

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